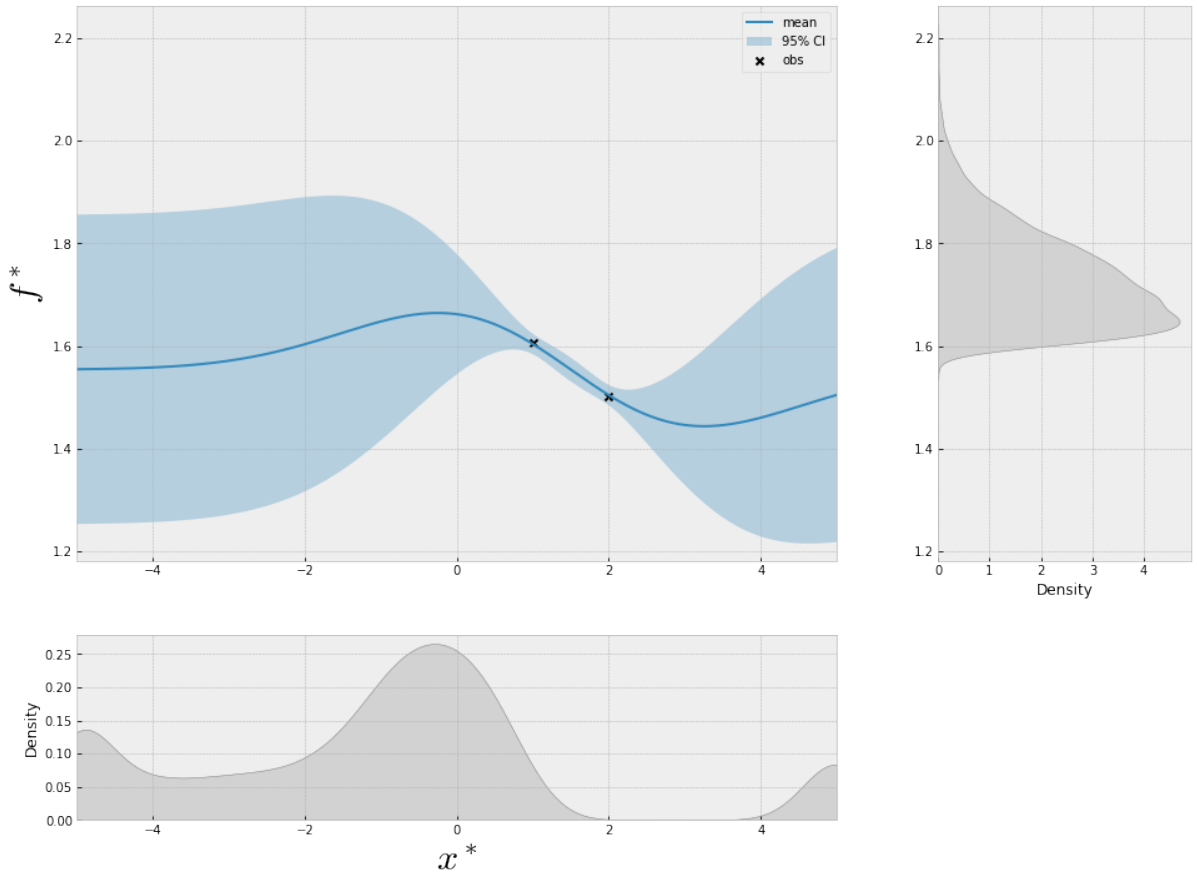


Figure 6.11 Gaussian process belief (top left), the distribution of the function optimizer x* (bottom), and the distribution of the optimal value f* (right). The distribution of the optimal value is always one-dimensional and therefore easier to work with than that of the optimizer.

An example of the distribution of the optimal value f^* induced by a GP is shown in Figure 6.11, on the right panel. We see that this distribution is roughly truncated around 1.6, which is exactly the value of the incumbent in our training data set; this makes sense as the optimal value f^* must be at least the incumbent value 1.6.

NOTE **The main advantage to focusing our effort on this distribution of f* is the fact that the distribution is always one-dimensional, regardless of the dimensionality of the objective function.**

Specifically, we can draw samples from this one-dimensional distribution to approximate the expected posterior entropy upon making a query. From this point, we follow the same idea behind entropy search: choosing the query that (approximately) minimizes the expected posterior entropy, or in other words, maximizes the expected *reduction* in entropy.

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